

Hamish Chhagan

Full Stack Developer

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PROJECTS

WriteLab — AI Content Generation & SEO Platform | 2025-2026

- Reduced content research time by 80% by consolidating 6+ page transitions into a single unified interface with parallel API orchestration using React 19 and TanStack Query.
- Enabled multi-model AI workflows by integrating 4 LLM providers (Claude, Gemini, DeepSeek, Perplexity) through LangGraph orchestration with automatic fallback handling.
- Improved content relevance by building RAG system with pgvector and OpenAI embeddings, enabling context-aware generation from crawled website content.
- Unified SEO data access by implementing single OAuth flow connecting Search Console, Analytics 4, and Business Profile with 7-day caching layer.
- Eliminated plugin security risks by developing WordPress publishing via Application Passwords, supporting draft/live publishing and Yoast/Rank Math metadata.
- Secured user data by implementing Passport.js authentication, CSRF/SSRF protection, rate limiting, and AES-256 encrypted credential storage.

E-Commerce Data Pipeline & Product Catalogue System | Oct 2025 – Feb 2026

Carrel Electrade — NZ electrical components distributor

- Processed 10,888 products across 169 columns by building end-to-end ETL pipeline that merged 3 data sources (PDF, web scraping, inventory) using Python and pandas.
- Achieved 98.8% image coverage from only 315 source images by designing intelligent series-level propagation logic (section → subsection → brand fallback).
- Extracted 4,829 products from 134-page PDF by engineering custom regex patterns with pdfplumber, handling malformed prices and inconsistent formatting.
- Automated product categorisation with 95%+ accuracy by integrating GPT-4o-mini with batch processing (20 products/call) and rule-based validation fallback.
- Enabled automatic variable product detection by engineering 27 SKU grouping patterns that identify varying attributes (voltage, current, colour, size).

Marketing Mailer — SaaS Email Marketing Platform | 2025

- Enabled high-volume email sending by building queue-based processing with BullMQ, supporting 10 emails/sec with 5 concurrent workers and automatic retry logic.
- Ensured tenant data isolation by designing multi-tenant architecture with user-provided API keys and AES-256 encrypted credential storage using Next.js 16 and Prisma.
- Provided real-time campaign insights by integrating TinyBird analytics via webhook ingestion, tracking delivery, opens, clicks, bounces, and complaints.
- Improved email deliverability by implementing DNS-verified domain management (SPF, DKIM, DMARC) with Resend API integration.

AI-Powered Lead Generation Chatbot Platform

- Enabled accurate knowledge-based responses by implementing RAG system with LangChain, FAISS vector store, and OpenAI embeddings for PDF knowledge bases.
- Automated lead capture by creating dynamic tool generation system with Zod schemas, allowing customisable data collection with automatic email notifications.
- Improved performance and type safety by migrating backend from Flask to Express.js with TypeScript, maintaining full feature parity.

The Phone Mason — Phone Repair Business Website

- Improved search position from 28 to 15 by implementing technical SEO, content optimisation, and local citation building over 16 months.
- Generated 920+ organic clicks and 150K+ impressions by achieving page 1 rankings for 'phone repairs' (#10), 'samsung battery replacement' (#10), and 'android repair' (#9).
- Created lead generation engine by building complete website from scratch with service pages, online booking, and contact forms optimised for local search.

Kiwi Kandy — WooCommerce E-commerce Platform

- Increased organic clicks by 445% and impressions by 115% by executing SEO strategy that improved average position from 51 to 12 over 6 months.
- Expanded sales channels by integrating Google Merchant Centre for automated product feeds, enabling visibility across Google Shopping.
- Reduced order processing time by developing shipping and fulfilment automation including label generation, inventory tracking, and customer notifications.

WORK EXPERIENCE

Fullstack Developer | August 2024 – Present

Rank / Web Merchants | Auckland CBD

- Processed 10,400+ products with 98.8% image coverage by building ETL pipeline for Carrel Electrade using Python, pandas, and pdfplumber.
- Streamlined compliance workflows by developing custom LMS for Test and Tag NZ with certification tracking and automated reporting.
- Improved user experience and conversions by leading full website redesign for SmashedIt.co.nz with modern UI/UX and payment integration.
- Enhanced purchase journey by creating custom product configurators and checkout improvements for The Cylinder Guy e-commerce platform.
- Enabled data-driven decisions by implementing Google Tag Manager and GA4 across client sites with custom event tracking and conversion funnels.
- Improved organic visibility by conducting SEO campaigns including technical audits, keyword strategy, and backlink acquisition for multiple clients.
- Solved complex scheduling requirements by developing WordPress/WooCommerce plugins for event booking with date management and seat allocation.

Business Development Consultant | March 2024 – August 2024

PENINSULA | Auckland CBD

- Improved client relationships by developing communication skills through direct stakeholder engagement and translating technical concepts for non-technical audiences.
- Gained commercial awareness by working in metrics-driven environment, learning to link technical deliverables to measurable business outcomes (KPIs).

Senior Repair Tech & Manager | July 2016 – Nov 2023

SMASHED IT / ILAB | Auckland

- Built capable technical team by managing and training staff in diagnostics, repairs, and operational procedures.
- Drove business growth by taking ownership of website management, SEO optimisation, and ad campaign creation.
- Reduced training time by authoring process guides and SOPs that standardised quality control procedures.

CORE COMPETENCIES & SKILLS

AI & Machine Learning: LangChain, LangGraph (Multi-Agent Systems), OpenAI API (GPT-4, GPT-4o, text-embedding-ada-002), Anthropic Claude API, Google Gemini API, DeepSeek, Perplexity, RAG (Retrieval-Augmented Generation), FAISS, pgvector, PDF Processing, Web Scraping, LLM Prompt Engineering, Chatbot Development, Semantic Search

Data Engineering: pandas, ETL Pipelines, pdfplumber, BeautifulSoup, Data Cleaning & Enrichment, Regex Pattern Engineering, Batch Processing

Frontend: React 19, TypeScript, JavaScript, Next.js 16, React Native, HTML5, CSS3, Tailwind CSS, Mantine UI, TanStack Query, Zustand, TipTap, Figma

Backend: Node.js, Express.js, Python, PHP, WordPress, WooCommerce Plugin Development, REST APIs, WebSockets, Prisma ORM, Sequelize ORM, Zod Validation, Session Management, Authentication & Authorisation

Databases: PostgreSQL (pgvector), MySQL, Redis, DynamoDB

DevOps & Cloud: Docker, Docker Compose, CI/CD, GitHub Actions, AWS (EC2, S3, RDS, Lambda), Google Cloud Platform, Linux Server Administration, SSH, Apache, NGINX, Proxmox VE

Automation & Integrations: n8n.io, BullMQ (Queue Processing), TinyBird (Analytics), Resend API, Nodemailer (SMTP), WooCommerce, Google Merchant Centre, Google Search Console, Google Analytics 4, Google Business Profile, Stripe, SEO Optimisation

EDUCATION

Bachelor of Science, Computer Science | Graduated Feb 2024

University of Auckland | Auckland, New Zealand